

The MOND interpolation function and flat rotation curves are mutually exclusive for minimally coupled k-essence

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Abstract

Relativistic formulations of MOND, from RAQUAL to TeVeS, couple the scalar field to matter through a conformal or disformal relation. This choice is standard but has not, to our knowledge, been justified from first principles in the literature. We show that, for a single k-essence scalar, it cannot be dropped. For a k-essence scalar with kinetic function $F(X)$ and *minimally coupled* matter—so that the scalar acts on test particles only through the back-reaction of $T_{\mu\nu}^\phi$ on the metric—the circular velocity in the static spherical exterior obeys $v^2 \propto r^{-2/(2n-1)}$ for $F = AX^n$. Flat curves are attained only in the limit $n \rightarrow \infty$, where the effective profile is $\rho_\phi \propto r^{-2}$, an isothermal dark-matter halo. The deep-MOND interpolation $\mu(x) \rightarrow x$ requires $n = 3/2$, for which $v^2 \propto \ln r/r$. The two limits are mutually exclusive, and the exclusion extends to any F whose infrared branch is $X^{3/2}$. We further show that demanding *exact* flatness for arbitrary F fixes $F(X) = A(\sqrt{X} - \lambda)^2$ uniquely, which carries a ghost for $X < \lambda^2$ and whose $\mu = F'$ diverges rather than vanishing as $x \rightarrow 0$. The result is not obvious: Armendariz-Picon and Lim obtained asymptotically flat curves from minimally coupled k-essence. We show that none of their configurations reaches the deep-MOND regime, and that their polytropic family reproduces our scaling law exactly, with the MOND value $n = 3/2$ sitting at the boundary where their curves cease to flatten. Our statement extends to non-canonical kinetic terms a no-go established by Fabri for the minimal canonical scalar, recovering that case as $n = 1$.

1 Introduction

MOND phenomenology at galactic scales is summarised by an interpolation function $\mu(x)$, $x = g/a_0$, satisfying $\mu(x) \rightarrow x$ for $x \ll 1$ and $\mu(x) \rightarrow 1$ for $x \gg 1$. Every relativistic completion proposed since RAQUAL [1] realises this by coupling a scalar field to matter: the physical metric entering the matter action is a conformal ($g_{\mu\nu} = e^{2\phi}\tilde{g}_{\mu\nu}$) or disformal transform of the Einstein metric, and the scalar equation consequently acquires a baryonic source. TeVeS [2] and the stratified constructions [3] elaborate the field content but retain that coupling. The review literature [4] presents this architecture as the starting point without arguing that it is forced.

Here we ask whether it is. We consider a k-essence scalar with matter *minimally* coupled, so that the scalar influences test particles only through the back-reaction of its stress-energy tensor on the metric, and we ask whether any kinetic function $F(X)$ can then reproduce both the deep-MOND interpolation and flat rotation curves.

The question is not rhetorical. Armendariz-Picon and Lim [5] studied exactly this setting—minimally coupled k-essence, back-reaction only—and exhibited configurations with asymptotically flat rotation curves. Read in isolation, their result suggests the coupling is dispensable.

We show that, within this class, it is not, and locate precisely where the two readings part company.

A partial result in this direction is due to Fabri [7], who proved that a static spherical *minimal canonical* massless scalar cannot supply the halo density, and who identified the non-canonical case as an open problem. We close that case.

2 Setup and hypotheses

We work in the Einstein frame with signature $(-, +, +, +)$ and units $c = 1$. The scalar sector is

$$S_\phi = - \int d^4x \sqrt{-g} N F(X), \quad X \equiv \frac{(\partial\phi)^2}{a_*^2}, \quad (1)$$

with $N > 0$ a constant of dimension M^4 and a_* a fixed acceleration scale, so that X is dimensionless and ϕ is dimensionless. The numerical value of a_* , and of the MOND scale a_0 , plays no role in any of the results below; for the status of the proposed numerical relation between a_0 and H_0 , including its dependence on the distance ladder, see [9]. We state the hypotheses explicitly; the reason for doing so is discussed in Sec. 6.

- H 1.** The scalar sector is (1), i.e. a k-essence Lagrangian depending on $\partial\phi$ only through X .
- H 2.** Matter is minimally coupled, $S_{\text{mat}} = S_{\text{mat}}[g_{\mu\nu}, \Psi]$. The scalar exerts no direct force on matter and acts only through $T_{\mu\nu}^\phi$.
- H 3.** The configuration is static, spherically symmetric, and weak-field.
- H 4.** Outside the baryonic source the scalar equation is homogeneous.
- H 5.** $F \in C^2$ with $F'(X) > 0$ on the domain of interest (absence of ghosts).
- H 6.** $W(X) \equiv F'(X)\sqrt{X}$ is monotonic, so that $X(r)$ is invertible. This holds whenever $F' > 0$ and $F'' \geq 0$.

3 Two lemmas

Lemma 1 (field profile). *Under H1–H4, the exterior field satisfies*

$$r^2 W(X(r)) = \frac{C}{a_*}, \quad W(X) \equiv F'(X)\sqrt{X}, \quad (2)$$

with C an integration constant.

Proof. The scalar field equation is $\nabla_\mu(F'(X)\partial^\mu\phi) = 0$ by H2 and H4. For a static radial profile this integrates to $r^2 F'(X)\phi' = \text{const}$, and $\phi' = a_*\sqrt{X}$ by the definition of X . $\square$

Remark. Equation (2) fixes the profile of the *field*. It does not fix the profile of the *potential*, and the distinction is the origin of the obstruction below.

Lemma 2 (active source). *The trace-adjusted source entering the Newtonian potential is*

$$\rho_\phi + 3\bar{p}_\phi = 2a_*^2 S(X), \quad S(X) \equiv XF'(X) - F(X). \quad (3)$$

Proof. From (1), $T_{\mu\nu} = 2F'\partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}a_*^2F$ (absorbing N into a_*^2). For a static radial profile, $\rho = a_*^2F$, $p_r = 2F'(\phi')^2 - a_*^2F$ and $p_t = -a_*^2F$, whence $3\bar{p} = p_r + 2p_t = 2F'(\phi')^2 - 3a_*^2F$. Adding and using $(\phi')^2 = a_*^2X$ gives (3). $\square$

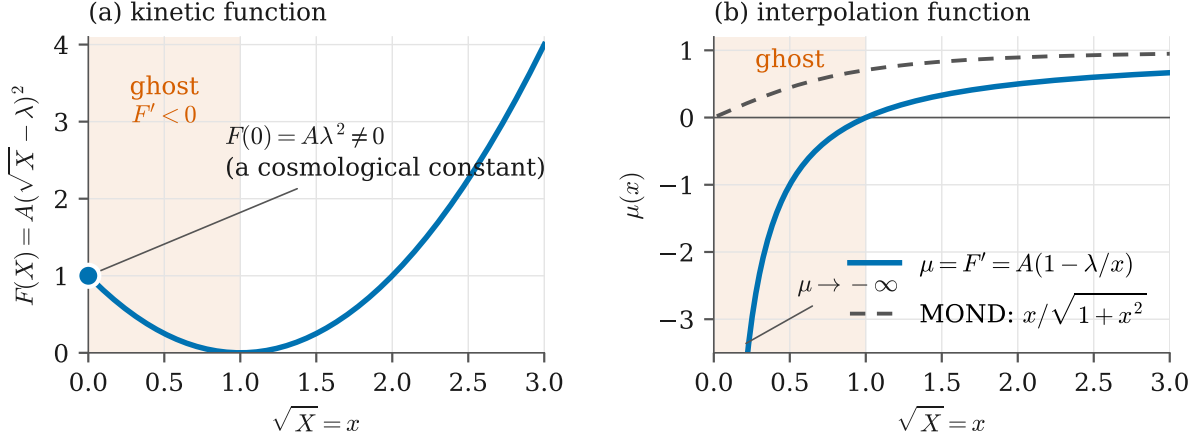


Figure 1: The unique exactly-flat solution of Theorem 1, $F = A(\sqrt{X} - \lambda)^2$ (shown for $A = \lambda = 1$). (a) $F(0) = A\lambda^2 \neq 0$ acts as a cosmological constant in the infrared. (b) $\mu = F'$ is negative (a ghost, shaded) for $x < \lambda$ and diverges to $-\infty$ as $x \rightarrow 0$, instead of following the deep-MOND behaviour $\mu \rightarrow x$ (dashed).

4 Exact flatness

Theorem 1 (unique solution, and it is sick). *Under H1–H6, exact flatness $v^2 = \text{const}$ holds if and only if*

$$F(X) = A(\sqrt{X} - \lambda)^2 \quad (4)$$

for constants $A > 0$, $\lambda > 0$. This solution has $F'(X) = A(1 - \lambda/\sqrt{X})$, hence a ghost for $X < \lambda^2$; it satisfies $F(0) = A\lambda^2 \neq 0$, a cosmological constant rather than a deep-MOND limit; and $\mu = F' \rightarrow -\infty$ as $x \rightarrow 0$.

Proof. Flatness requires $\nabla^2 \Psi \propto r^{-2}$, hence $S(X(r)) \propto r^{-2}$ by Lemma 2. By Lemma 1, $W(X(r)) \propto r^{-2}$ as well. By H6, $X(r)$ is invertible, so $S(X) = \lambda W(X)$ as functions, i.e. $XF' - F = \lambda F' \sqrt{X}$. This first-order equation integrates to the stated form; the listed properties follow by direct differentiation (Fig. 1). The degenerate case $\lambda = 0$ gives $F = AX$ and $S \equiv 0$: the source vanishes identically and no configuration is sourced at all. $\square$

5 The main result

Theorem 2 (mutual exclusion). *Let $F = AX^n$ under H1–H6. Then*

$$\rho_\phi \propto r^{-4n/(2n-1)}, \quad v^2 \propto r^{-2/(2n-1)} \quad (n > 3/2). \quad (5)$$

For $1/2 < n < 3/2$ the scalar's enclosed mass converges and $v^2 \propto r^{-1}$; at $n = 3/2$, $v^2 \propto \ln r/r$. Consequently:

- (i) No finite n gives an exactly flat curve; flatness is approached only as $n \rightarrow \infty$, where $\rho_\phi \rightarrow r^{-2}$, the isothermal profile.
- (ii) The deep-MOND interpolation $\mu(x) = F'(x^2) \rightarrow x$ as $x \rightarrow 0$ requires $n = 3/2$ uniquely, for which $v^2 \propto \ln r/r$.

Hence the MOND interpolation function and flat rotation curves cannot be obtained simultaneously.

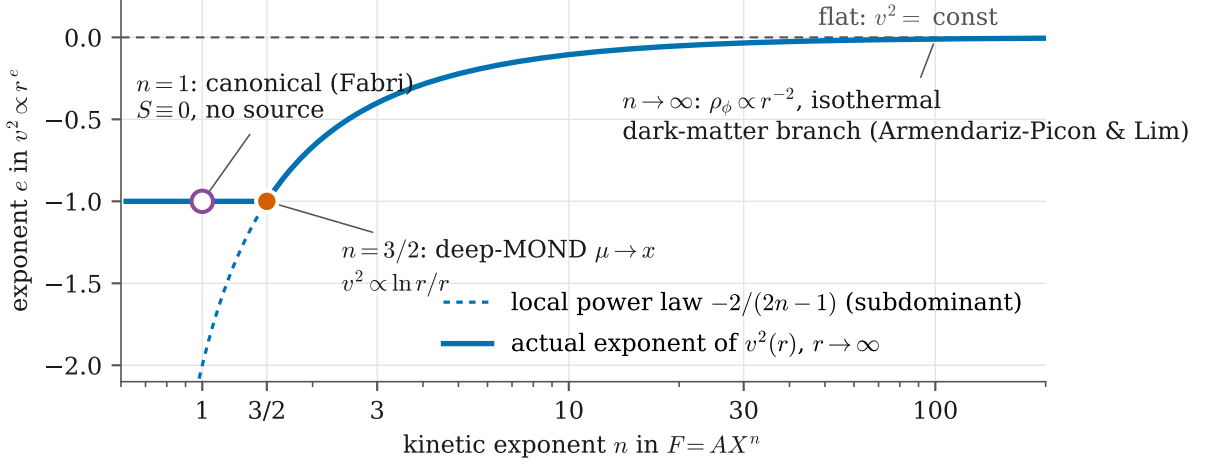


Figure 2: Asymptotic exponent e of $v^2 \propto r^e$ for $F = AX^n$ under minimal coupling (Theorem 2). For $n > 3/2$ the halo dominates and $e = -2/(2n - 1)$; below $n = 3/2$ the scalar's enclosed mass converges and the curve is Keplerian (the local power law, dotted, is subdominant). The deep-MOND value $n = 3/2$ sits at the marginal point, $v^2 \propto \ln r/r$. Flatness ($e = 0$) is reached only as $n \rightarrow \infty$, the barotropic branch of [5]. The canonical case $n = 1$ [7] has vanishing active source.

Proof. For $F = AX^n$, Lemma 1 gives $W = nAX^{n-1/2}$, so $X \propto r^{-4/(2n-1)}$. Lemma 2 gives $S = (n - 1)AX^n$, so $\nabla^2 \Psi \propto r^{-p}$ with $p = 4n/(2n - 1)$. The enclosed active mass is $M(< r) \propto \int^r r'^{2-p} dr'$, which grows as r^{3-p} when $p < 3$ ($n > 3/2$), as $\ln r$ when $p = 3$ ($n = 3/2$), and converges when $p > 3$ ($1/2 < n < 3/2$). With $v^2 = GM(< r)/r$ this gives $v^2 \propto r^{2-p} = r^{-2/(2n-1)}$ for $n > 3/2$, which is (5); $v^2 \propto \ln r/r$ at $n = 3/2$; and the Keplerian $v^2 \propto r^{-1}$ below. Setting the exponent in (5) to zero gives $4n = 4n - 2$, with no solution, and the other two branches decrease, proving (i); the limit is immediate. For (ii), $\mu = F' = nAX^{n-1} \propto x^{2n-2}$, and $\mu \propto x$ forces $n = 3/2$, the marginal case $v^2 \propto \ln r/r$ (Figs. 2 and 3). $\square$

Remark. The restriction $n > 1/2$ is not accidental. At $n = 1/2$, W is constant and Eq. (2) admits no non-trivial exterior solution. For $n < 1/2$, Lemma 1 gives $X \propto r^{4/(1-2n)}$, which grows without bound at large r , violating H3; the active source $S = (n - 1)AX^n$ is moreover negative. The value $n = 3/2$ selected by $\mu \rightarrow x$ lies well inside the admissible range.

The value $n = 3/2$ is not an arbitrary choice imposed on the class. It is the infrared kinetic branch of RAQUAL [1] and of the constructions built on it [3, 2]: the deep-MOND limit $\mu(x) \rightarrow x$ requires $F'(X) \propto X^{1/2}$, hence $F \propto X^{3/2}$ in the infrared. In those theories this branch produces the MOND force because matter couples to the scalar directly. Under minimal coupling, Theorem 2 shows that the same branch is precisely what prevents the back-reaction from producing a flat curve. The conclusion does not rest on the power-law form:

Corollary 1. *Let $F = AX^{3/2} + G(X)$ satisfy H1–H6, with $G(X)$ and $XG'(X)$ both $o(X^{3/2})$ as $X \rightarrow 0$; equivalently, $\mu(x) \rightarrow x$ in the deep-MOND limit. Then $v^2 \propto \ln r/r$ at large r . No kinetic function in the class yields both the MOND interpolation and a flat rotation curve.*

Proof. Since $F' = \frac{3}{2}AX^{1/2}[1 + o(1)]$, Lemma 1 gives $X \simeq 2C/(3Aa_*r^2) \rightarrow 0$ in the exterior, so the asymptotics are governed by the infrared branch. Lemma 2 then gives $S = \frac{1}{2}AX^{3/2}[1 + o(1)] \propto r^{-3}$; the enclosed active mass grows as $\ln r$ and $v^2 \propto \ln r/r$. A term linear in X , which contributes nothing to S , would dominate F' as $X \rightarrow 0$ and violate $\mu \rightarrow x$. $\square$

Remark. Setting $n = 1$ in (5) gives $\rho_\phi \propto r^{-4}$, recovering the canonical result of [7]. We note in passing that the obstruction in that case is in fact stronger than stated there: by Lemma 2,

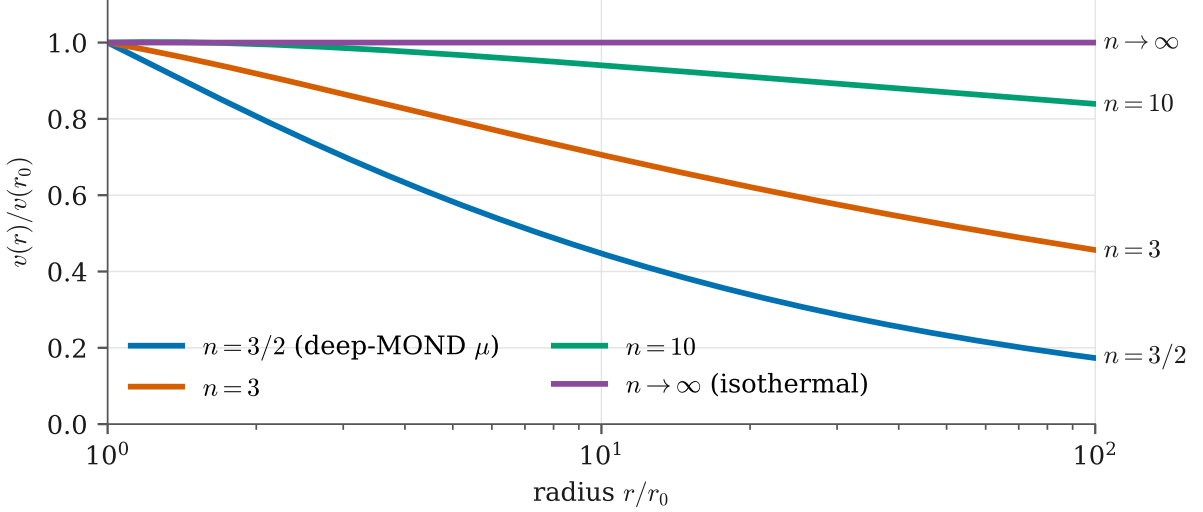


Figure 3: Rotation curves of the scalar halo, $v^2 = GM(< r)/r$ with the enclosed mass integrated from the profile $\rho_\phi \propto r^{-4n/(2n-1)}$ (inner cutoff $0.1 r_0$), normalised at r_0 . Only the $n \rightarrow \infty$ limit is flat; the value selected by the MOND interpolation function, $n = 3/2$, gives the steepest decline of the four.

$S(X) = 0$ identically at $n = 1$, so the active source of the Newtonian potential vanishes and the massless canonical scalar does not gravitate at all in this limit, rather than gravitating with the wrong power.

6 Relation to previous work

Armendariz-Picon and Lim [5] studied static, spherically symmetric configurations of minimally coupled k-essence—the setting of H1–H4—and obtained flat or asymptotically flat rotation curves in three families. Their results are correct, and none of them lies on the MOND branch.

For barotropic Lagrangians $F \propto X^n$ the curve flattens as $n \rightarrow \infty$, where $\rho_\phi \propto r^{-2}$: the isothermal dark-matter branch of Theorem 2, with $\mu \propto X^{n-1}$.

For the generalised Chaplygin family (their polytropic case with $p_* < 0$, $\gamma < 0$), the Lagrangian behaves near $X = 0$ as a constant plus a term $\propto X^{(1-\gamma)/2}$, i.e. as $F = AX^n$ with $n = (1 - \gamma)/2$. Their rotation curve, $v^2 \propto r^{2/\gamma}$ for $\gamma < -2$ and $v^2 \propto r^{-1}$ for $\gamma > -2$, is then exactly Eq. (5) together with the Keplerian regime below $n = 3/2$. This is an independent check of Theorem 2 against a published calculation. The MOND value $n = 3/2$ corresponds to $\gamma = -2$, precisely the boundary of their two regimes, where the curve does not flatten ($v^2 \propto \ln r/r$); flatness requires $\gamma \rightarrow -\infty$.

For polytropic Lagrangians with $p_* > 0$, $0 < \gamma < 1$ —still a pure, shift-symmetric $L(X)$ with no potential, i.e. within class (1)—the curve is asymptotically flat and the configuration is stable, as they show, but the support is the radial pressure rather than the density: the field gradient tends to a *non-zero* constant, $X \rightarrow X_c$, so that ϕ grows linearly in r ; X_c is the zero of the Lagrangian, at which also $F'(X_c) = 0$, so the energy density $\rho_\phi \propto F(X_c)$ vanishes without $F(0)$ entering; then $r^2 p_r \rightarrow \text{const}$ while $r^2 \rho_\phi \rightarrow 0$. This is the healthy asymptotic counterpart of the exact solution of Theorem 1, which likewise approaches $X = \lambda^2$ with $F'(\lambda^2) = 0$: the ghost there is the price of *exact* flatness, not of flatness as such. Moreover, this Lagrangian is not defined for $|X| < |X_c|$, so for it a deep-MOND limit $\mu(x) \rightarrow x$ as $x \rightarrow 0$ does not exist at all.

What Theorem 2 and Corollary 1 add is the converse: the branch that does reach the deep-MOND regime gives $v^2 \propto \ln r/r$. The two limits are mutually exclusive.

Fabri [7] established the corresponding statement for a minimal canonical scalar and listed the non-canonical kinetic term as an explicit open gateway. Theorem 2 closes that gateway, and recovers the canonical case as $n = 1$.

Finally, a disclosure. The question arose in the Λ -EG framework [10], whose kinetic function has the infrared branch $F = \frac{2}{3}X^{3/2}$ and whose matter sector is stated to be minimally coupled. Within those stated hypotheses, Theorem 2 and Corollary 1 apply: the scalar back-reaction cannot produce flat rotation curves. The flat curves obtained in [10] come instead from an effective parametrisation $\Phi = \Phi_N + c^2\alpha\phi$, which, written as a varied action, is a conformal coupling $A^2(\phi)g_{\mu\nu}$ and lies outside the class considered here. Its Open Problem O9—the derivation of the non-relativistic limit from $T_{\mu\nu}^\phi$ alone—is therefore not closable within minimally coupled k-essence. The theorem itself is independent of Λ -EG and would stand were that framework falsified.

Bruneton and Esposito-Farèse [6] caution that published no-go theorems in this context often rest on hypotheses that may be unnecessary, and that the space of viable models is therefore larger than such theorems suggest. We have accordingly stated H1–H6 explicitly. The only non-obvious one is H6, the monotonicity of $W = F'\sqrt{X}$, which is required for $X(r)$ to be invertible in the proof of Theorem 1; it is implied by the standard no-ghost condition $F' > 0$ together with convexity $F'' \geq 0$, and is therefore not an additional physical restriction within the class usually considered. Nor is it needed for the main conclusion: if W is not monotonic, Eq. (2) has several branches, each approaching a distinct zero of W as $r \rightarrow \infty$; only the branch with $X \rightarrow 0$ probes the deep-MOND regime, and Corollary 1 applies to it.

7 Discussion

The content of Theorem 2 can be read as an explanation of an architectural choice that the relativistic-MOND literature has made uniformly without arguing for it. In RAQUAL [1] the scalar equation carries a baryonic source because matter couples to $e^{2\phi}\tilde{g}_{\mu\nu}$; the physical potential is then $\Phi = \Phi_N + c^2\phi$, and the MOND force is the gradient of the scalar itself rather than the gravitational response to its energy density. Under minimal coupling that channel is closed, and only the back-reaction remains—which, by (5), delivers the wrong radial profile in precisely the regime where the MOND interpolation is defined.

The same reading applies to the subsequent development of the field. Vector degrees of freedom were introduced [3, 2] to repair the lensing behaviour of the conformal coupling, not to restore a force that minimal coupling had removed; the coupling itself was never in question. Theorem 2 states what would have been lost had it been dropped.

The result is complementary to a geometric one. The standard interpolating function $\mu(x) = x/\sqrt{1+x^2}$ admits an exact characterisation as the cumulative distribution of $\cot\theta$ for directions distributed isotropically on the hemisphere [8]. That statement concerns the *form* of μ and is silent on the field theory that realises it. Theorem 2 and Corollary 1 constrain the latter: whatever the origin of μ , its deep-MOND branch cannot be carried by the back-reaction of a minimally coupled k-essence field, and requires matter to couple to the scalar directly.

Two limitations should be stated. First, the result concerns the static spherical exterior; it does not address time-dependent configurations, where a disformal coupling can contribute. Second, it is a statement about the class (1): theories with additional fields, with non-minimal curvature couplings, or with screening mechanisms that render F environment dependent are outside its scope, as are the evasion routes catalogued in [6].

Data and code availability

The script `DOS_TEOREMAS_2026-09-24.py` provides a self-contained symbolic verification of Lemmas 1–2, Theorems 1–2 (including the marginal case $n = 3/2$), Corollary 1, the degenerate case $\lambda = 0$, and the consistency checks against [5] (including their Eq. (45) and polytropic cases) and [7]; all 31 checks pass. Figures 1–3 are generated by `figuras_nogo.py`.

References

- [1] J. Bekenstein and M. Milgrom, *Does the missing mass problem signal the breakdown of Newtonian gravity?*, *Astrophys. J.* **286**, 7 (1984).
- [2] J. D. Bekenstein, *Relativistic gravitation theory for the modified Newtonian dynamics paradigm*, *Phys. Rev. D* **70**, 083509 (2004).
- [3] R. H. Sanders, *A stratified framework for scalar-tensor theories of modified dynamics*, *Astrophys. J.* **480**, 492 (1997).
- [4] B. Famaey and S. S. McGaugh, *Modified Newtonian Dynamics (MOND): observational phenomenology and relativistic extensions*, *Living Rev. Relativity* **15**, 10 (2012).
- [5] C. Armendariz-Picon and E. A. Lim, *Haloes of k-essence*, *JCAP* **0508**, 007 (2005), [arXiv:astro-ph/0505207](https://arxiv.org/abs/astro-ph/0505207).
- [6] J.-P. Bruneton and G. Esposito-Farèse, *Field-theoretical formulations of MOND-like gravity*, *Phys. Rev. D* **76**, 124012 (2007), [arXiv:0705.4043](https://arxiv.org/abs/0705.4043).
- [7] D. D. Fabri, *Ray Time Theory (RTT) v2.6: trace-coupled scalar sector, barotropic-fluid closure, and a minimal-canonical no-go theorem*, Zenodo preprint (2026), [doi:10.5281/zenodo.20301769](https://doi.org/10.5281/zenodo.20301769).
- [8] J. P. Figueroa Torres, *A geometric characterization of the standard MOND interpolating function: $\mu(x) = x/\sqrt{1+x^2}$ as the isotropic distribution of $\cot \theta$* , preprint, in *Λ -EG — Emergent Gravitational Coherence: Theory, Ontology, Cosmology (T.O.C.)*, Zenodo (2026), [doi:10.5281/zenodo.22216334](https://doi.org/10.5281/zenodo.22216334).
- [9] J. P. Figueroa Torres, *Λ -EG $\rightarrow a_0$ - H_0 : a consistency test and its distance-ladder circularity*, Zenodo preprint (2026), [doi:10.5281/zenodo.21764888](https://doi.org/10.5281/zenodo.21764888).
- [10] J. P. Figueroa Torres, *Λ -EG Emergent Gravitational Coherence and Galactic Dynamics Without Particle Dark Matter*, v10, Zenodo preprint (2026), [doi:10.5281/zenodo.22166062](https://doi.org/10.5281/zenodo.22166062).